An Integrated QKD–PQC Framework for Secure Iris-Based Biometric Authentication System

***Abstract*—As cybersecurity evolves, biometric authentication offers a secure, user-centric approach to identity verification. However, these systems depend on reusable biometric templates, whose compromise permanently exposes the identity. This paper proposes an end-to-end framework for iris biometric authentication, avoiding reusable templates. It integrates iris verification, post-quantum authentication, and quantum-secured session key establishment seamlessly. The proposed architecture employs local iris reconstruction, eliminating the need for reusable biometric templates. It uses ML-DSA for secure terminal-server mutual authentication and signed QAuth session authorization. T12 biased-basis quantum key distribution (QKD) establishes quantum-secured keys across the communication channel securely. Recursive cascade reconciliation corrects errors, followed by privacy amplification for enhanced key security. HKDF-based derivation generates session keys, while AES-GCM protects subsequent communication between authenticated parties. We study the impact of communication distance, basis bias, and adversary interception scenarios of QKD on the performance of the proposed system. Our simulation results demonstrate that high legitimate-session reliability is achievable up to distances of approximately 150 km under the validated baseline configuration.**

***Index Terms*—Post-Quantum Cryptography, Quantum Key Distribution, Iris Biometrics, Biometric Authentication, Biometric Reconstruction**

I. Introduction

Biometric authentication has become a foundation of modern digital identity systems [1]. Security-sensitive domains have started to replace conventional credentials such as passwords and cards with biometric authentication. Today its use is extended to banking, healthcare, border security, and national digital infrastructure where reliable identification is mandatory [2]. Biometric features such as fingerprints, facial features, and iris patterns are especially useful because they are closely related to an individual. Traditional biometric systems commonly retain digital representations of biometric information that are subsequently required during identity matching. If this stored information gets compromised, it leaks the user identity forever. Protecting the biometric component alone, however, does not secure the complete authentication environment. Beyond biometric protection, communication security must address quantum threats that endanger Rivest–Shamir–Adleman (RSA) and Elliptic Curve Cryptography (ECC), motivating Post-Quantum Cryptography (PQC) [3]. Fuzzy extractor techniques offer a potential alternative by deriving consistent cryptographic secrets from inherently noisy biometric measurements while accommodating natural variations during acquisition [4]. The Module-Lattice Digital Signature Algorithm (ML-DSA) addresses this requirement by providing quantum-resistant digital authentication, while quantum key distribution (QKD) offers a different capability through the establishment of cryptographic keys based on quantum mechanical principles [5]. However, QKD requires an authenticated classical channel to prevent man-in-the-middle attacks, while PQC alone does not provide secure key distribution. These limitations and capabilities make PQC and QKD complementary technologies within a broader security architecture. Furthermore, practical implementations must account for factors beyond the individual cryptographic components, including optical attenuation, detector inefficiency, reconciliation leakage, and communication latency of QKD. Integrating these requirements into one coherent authentication pipeline therefore remains a significant research challenge [6].Existing approaches rarely integrate template-less biometrics, quantum-resistant authentication, and authenticated quantum key establishment end-to-end.

To address the existing gaps [7], [8], this paper presents an integrated post-quantum biometric authentication framework that brings these components together within a unified protocol pipeline. The proposed architecture combines biometric key reconstruction based on fuzzy extraction principles with ML-DSA-based mutual authentication. It additionally incorporates quantum authorization (QAuth) session tokens and T12 biased-basis QKD for session establishment [9]. Recursive cascade reconciliation is employed for error correction, followed by privacy amplification and HMAC-based Extract-and-Expand Key Derivation Function (HKDF-based) session key derivation. The resulting session keys are subsequently used to protect application-layer communication through Advanced Encryption Standard in Galois Counter Mode (AES-GCM). The complete framework is implemented as a functional research prototype with a graphical interface supporting end-to-end execution. Its performance is evaluated under realistic physical-layer conditions involving optical attenuation, detector efficiency, dark counts, and distance-dependent communication latency. The remainder of this paper details the proposed architecture and its implementation methodology. The subsequent sections discuss biometric reconstruction, post-quantum authentication, quantum key establishment, secure communication, and experimental evaluation.

II. Related Work

The following discussion in this section progressively examines the four overlapping directions of biometric template protection, post-quantum authentication, quantum secured communication, and hybrid architectures combining them.

A. Biometric Systems

Biometric credentials cannot be replaced because they are inherently associated with an individual. Therefore, maintaining confidentiality during the storage and transmission of biometric templates is essential [10]. Approaches such as cancellable biometrics, fuzzy commitments, and fuzzy extractors have been proposed to address these concerns, with fuzzy extractors deriving reproducible cryptographic material from noisy biometric data using non-secret helper information [11]. The integration of fuzzy commitments with blockchain has further introduced decentralized approaches for biometric protection [12]. However, these approaches primarily address biometric security and do not resolve quantum vulnerabilities affecting the surrounding authentication and communication infrastructure.

B. PQC-Based Authentication

According to NIST standardised ML-DSA approach is used to authenticate the digital signature [13]. Bauspieß *et al.* demonstrated post quantum secure two part computation for iris template protection [14]. However, PQC authenticates computationally, while quantum communication establishes keys with information-theoretic security. It creates the need for complementary architectures where PQC guarantees identity and quantum communication establishes a secure secret key.

C. Quantum Communication

QKD allows two parties to establish a secret key while interception in between is observable through statistical disturbance. BB84 establishes the fundamental prepare-and-measure model for QKD [15] while Lo *et al.* later showed improvements in sifting efficiency through asymmetric basis selection [16]. Practical QKD further needs classical post-processing. It includes reconciliation through mechanisms like Cascade [17] and privacy amplification to reduce the adversarial information [18]. However, QKD also requires biometric identity verification and authentication on the classical communication channel.

D. Current Hybrid Systems

Recent works like Qasim *et al.* [7] have begun integrating PQC, quantum communication, and biometrics. They combine ML-KEM, QSDC inspired mutual authentication, and QKD derived keys to encrypt and transmit biometric templates [7]. However, their architecture still retains the biometric template as a transmitted object.

III. Proposed System Architecture

The proposed architecture integrates biometric authentication, PQC and QKD into a single end-to-end authentication and secure communication pipeline as discussed below. Fig. 1 illustrates the proposed authentication framework, comprising user enrollment and authentication phases. Enrollment occurs once, whereas authentication is performed whenever users request secure communication access.

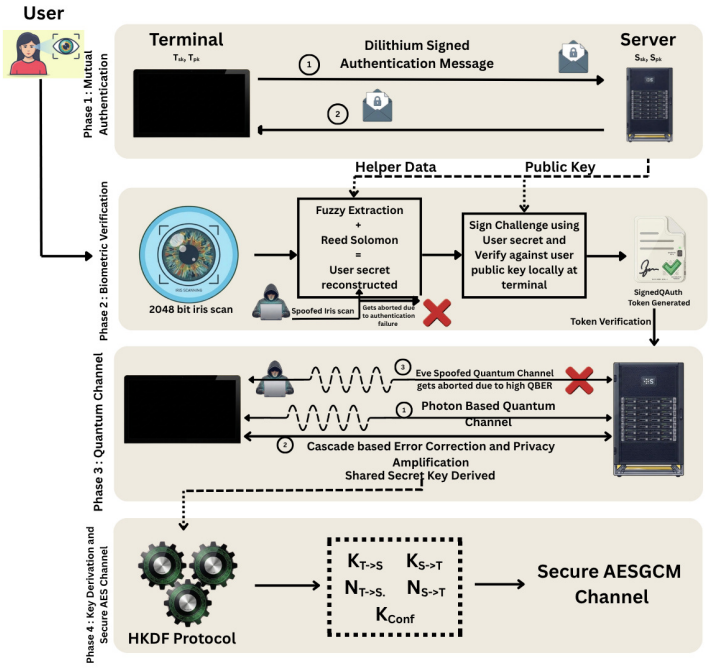


Fig. 1: Overall System Architecture for QBioAuth Pipeline

A. System Initialisation and Terminal-Server Mutual Authentication

During initialization, separate terminal and server entities are created with isolated memory spaces. Each node receives an ML-DSA public-private key pair and communicates only through predefined messages. Every session begins with mutual authentication before any user-specific information is processed. Mutual authentication proceeds as follows:

1) The terminal generates an authentication message containing timestamps and a fresh nonce, signs it using sk_T, and sends
$$\sigma_T = \mathrm{Sign}_{sk_T}(m \parallel t \parallel n) \tag{1}$$
where m is the authentication message, t is the issue times tamp and n is the nonce of the signature.
2) The server verifies the signature using pk_T,
$$\mathrm{Verify}_{pk_T}(m, \sigma_T) = 1 \tag{2}$$
and validates the timestamp and nonce for freshness and replay resistance.
3) The server performs the same signed challenge procedure in reverse, which the terminal verifies using the server's public key

Only after both verifications succeed is the endpoints considered mutually authenticated, permitting subsequent biometric enrollment or authentication.

B. User Enrollment

The enrollment process is performed once for each user and creates a long-term cryptographic identity associated with

the user's biometrics without creating, storing or transmitting a reusable biometric template. For an unregistered user, the terminal performs the following:

1) The user's unique identifier (email) and iris scan are acquired, and the iris scan is converted into a stable binary representation for fuzzy construction.
2) The fuzzy construction process derives stable data from which the user's ML-DSA key pair is deterministically generated.
3) The public verification key and non-secret helper data required for future biometric reconstruction are securely transmitted to the authentication server and associated with the user's identity.
4) After successful enrollment, the terminal immediately discards temporary enrollment data, including the iris scan, while the server retains only the public key and helper data.

Since the helper data alone can't reconstruct the enrolled biometric, server-side compromise will not expose the user's iris information or private credential.

C. User Authentication

For an enrolled user, authentication reconstructs the biometric-derived signing capability from a fresh iris observation without transmitting the biometric or private-key material. For user authentication, the terminal performs the following:

1) Using the user's email, the terminal retrieves the associated public verification key and helper data from the authentication server.
2) A fresh iris scan is converted into its binary representation and combined with the helper data through fuzzy reconstruction, tolerating limited acquisition noise.
3) Successful reconstruction locally regenerates the user's ML-DSA private signing key within the trusted terminal.
4) The reconstructed key signs a fresh authentication challenge, whose signature is verified against the retrieved public key.

Successful verification confirms the user's biometric identity without storing or transmitting the biometric template or private key. The protocol then proceeds to QAuth generation for quantum-session authorization.

D. QAuth Token Generation

Following a successful user authentication, the terminal generates a signed QAuth token that conveys cryptographic evidence of authentication to the server before quantum resources are allocated.

1) The terminal constructs
$$QAuth = \{UID, SID, T_{\text{issue}}, T_{\text{exp}}, Status\} \quad (3)$$
containing the authenticated user id (UID), session ID (SID), issue (T_{issue})and expiration (T_{exp}) timestamps, and authorization $Status$.
2) The token is signed using the terminal's private key,
$$\sigma_Q = \text{Sign}_{sk_T}(QAuth), \quad (4)$$
and transmitted to the server.
3) The server verifies the signature, token integrity and timestamp validity before authorising QKD session establishment.

QAuth therefore bridges biometric verification and quantum communication, ensuring that quantum resources are allocated only to authenticated terminals acting for successfully verified users.

E. Quantum Key Distribution for Session Establishment

Following successful QAuth verification, the server initiates T12 biased-basis QKD with the authenticated terminal to establish fresh shared secret material. The protocol proceeds as follows:

1) The terminal and server select their preparation and measurement bases according to
$$P(Z) = p, \qquad P(X) = 1 - p \quad (5)$$
with $p = 0.8$ in the validated baseline configuration.
2) Quantum states are transmitted through the distance-aware channel model incorporating fibre attenuation, detector inefficiency, intrinsic optical error, and dark counts. Fibre transmission follows
$$\eta_{\text{fiber}}(d) = 10^{-\alpha d} \quad (6)$$
where α the attenuation coefficient (in dB/Km) and d is the length of the optical fiber. Transmission continues until the required detector events satisfy
$$N_{\text{det}} \geq N_{\text{target}} \quad (7)$$
3) Both parties exchange signed basis information over the authenticated classical channel and discard mismatched-basis measurements.
4) A fixed fraction of the sifted key is disclosed for parameter and Quantum Bit Error Rate (QBER) estimation, with
$$QBER = \frac{N_{\text{error}}}{N_{\text{compared}}} \quad (8)$$
where N_{error}, N_{compared} denotes the number of mismatched bits and the number of disclosed bits, respectively, used for parameter estimation. Sessions exceeding the configured QBER threshold are aborted and a fresh quantum transmission is attempted.

The remaining sifted key is shared but may contain residual disagreements caused by channel and detector imperfections. Therefore, the sifted bits proceed to the information reconciliation step before cryptographic use.

F. Cascade-Based Information Reconciliation

The sifted keys may contain residual disagreements due to channel and detector noise and therefore cannot yet be used as shared cryptographic material. The framework applies Cascade-based reconciliation over the authenticated classical channel as follows:

1) The sifted key is partitioned into blocks B_i, and each party computes

$$P(B_i) = \bigoplus_{j=1}^{n_i} b_j \tag{9}$$

2) For blocks satisfying

$$P_T(B_i) \neq P_S(B_i) \tag{10}$$

binary search, recursively divides the block to identify and correct the erroneous bits. P_T and P_S are the parity blocks at the terminal and server sides.

3) Multiple reconciliation passes with recursive look-back correct residual disagreements missed in earlier passes until

$$K_T = K_S \tag{11}$$

where K_T and K_S denote the reconciled keys obtained by the terminal and the authentication server, respectively. Reconciliation produces identical terminal and server keys but publicly reveals parity information. The reconciled secret therefore undergoes privacy amplification before cryptographic use.

G. Privacy Amplification

To reduce information potentially exposed through parameter estimation, reconciliation, or quantum-channel observation, the reconciled secret is compressed through privacy amplification:

1) The reconciled secret is hashed as

$$K_{PA} = \mathrm{SHA256}(K_{REC}) \tag{12}$$

2) The retained secret length is determined by

$$L_{PA} = L_{REC} - \lambda \tag{13}$$

where L_{REC} denotes the reconciled key length and λ is the configured security margin, accounting for information leakage.

3) Both endpoints independently obtain the identical privacy-amplified secret,

$$K_{PA} = K_{PA}^{(T)} = K_{PA}^{(S)} \tag{14}$$

The resulting shared secret is subsequently passed to HKDF to derive independent cryptographic material for application-layer communication.

H. HKDF-Based Key Derivation and Classical Channel Communication

The privacy-amplified secret is expanded through HKDF into independent, purpose-specific session material:

$$\{K_{TS}, K_{ST}, N_{TS}, N_{ST}, K_{CONF}\} = \mathrm{HKDF}(K_{PA}, Salt, Info), \tag{15}$$

where K_{TS} and K_{ST} are direction-specific encryption keys, N_{TS} and N_{ST} are the corresponding nonce bases, and K_{CONF} confirms that both endpoints derived identical key material. The $Info$ is the quantum session identifier and $Salt$ is a random value appended to the secret material to enhance entropy.

Application-layer communication then proceeds as follows:

1) Terminal-to-server and server-to-terminal messages use their independently derived keys and nonce material.
2) AES-GCM provides authenticated encryption for each message,

$$(C, \tau) = \text{AES-GCM}_{K_{enc}}(M), \tag{16}$$

producing ciphertext C and authentication tag τ from plaintext M.

The resulting channel provides confidential and authenticated application-layer communication using session material derived from the reconciled and privacy-amplified quantum secret.

I. System Assumptions and Limitations

The framework assumes trusted enrollment and terminal execution, while biometrics, quantum transmission and detector behaviour are simulated rather than hardware-derived. Hardware side channels, physical terminal compromise, malicious insiders and denial-of-service attacks remain outside the threat model. Consequently, the reported distance and reliability results characterize the simulated configuration and should not be interpreted as hardware performance limits.

IV. Experimental Methodology

The proposed framework was evaluated step by step in a controlled simulation experiment environment to identify the effects and trustworthiness of the framework when physical conditions, quantum channel conditions and noises are varied.

A. Experimental Setup

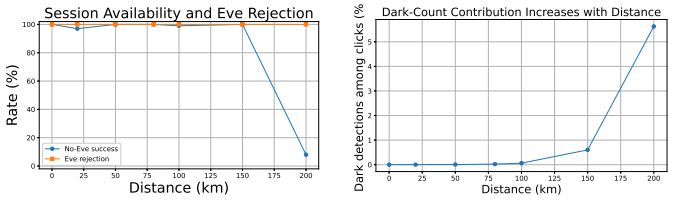
According to the implementation details mentioned in the Section III, a complete experiment study was conducted. Unless varied as a part of the simulation testing or experiment factor, the protocol and physical channel parameters are described in Table I. This creates a common operation ground from which the individual effects of selected parameters can be measured.

B. One-Factor-at-a-Time Experimental Study

A One-Factor-at-a-Time (OFAT) study varied one parameter while holding all others at the baseline in Table I, enabling isolation of its effects on security, reliability and resource efficiency. Three parameters of communication distance, Eve interception, and Z-basis probability spanning 18 experimental cases were evaluated, covering physical-channel condition, QKD configuration, and adversarial interception. A fixed master seed ensured reproducibility. Each case was evaluated over 100 trials without Eve and 100 trials under 100% interception, except the Eve-interception-fraction sweep itself, resulting in 3000 individual experimental records.

TABLE I
Baseline Configuration of the Proposed Framework

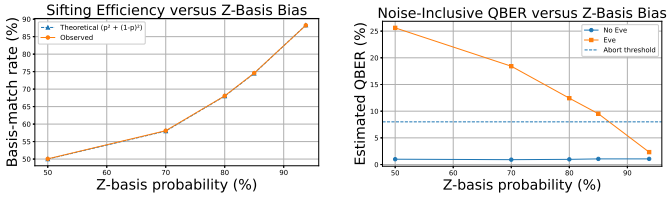
Parameter	Assumption / Role	Baseline
Z-basis probability	Probability of selecting the dominant T12 basis	80%
Parameter-estimation fraction	Fraction of sifted bits disclosed for QBER estimation	15%
QBER abort threshold	Maximum estimated QBER permitted before session rejection	8%
Target detector events	Minimum accumulated detections before QKD processing	8192
Transmission batch size	Optical pulses represented in each acquisition batch	8192
Source emission efficiency	Probability of successful photon emission	100%
Coupling efficiency	Efficiency of coupling photons into the optical channel	80%
Detector efficiency	Mean photon detection efficiency	20%
Intrinsic optical error	Mean baseline optical error probability	1%
Dark-count probability	Probability of a spurious detector event per gate	10^{-6}
Dark-count error probability	Probability that a dark-count detection produces an incorrect bit	50%
Cascade passes	Configured recursive reconciliation passes	6
Privacy security margin	Additional key-length deduction during privacy amplification	128 bits
Maximum QKD attempts	Maximum fresh QKD attempts following recoverable failure	3
Synthetic iris noise	Bit-error probability applied to a repeated synthetic iris scan	1%
Synthetic iris rotation	Circular displacement used to model acquisition misalignment	-4 bits
QAuth validity	Maximum authorisation-token lifetime	120 s



(a) Effect of Distance on Session Success and Acceptance

(b) Effect of Distance on Dark Counts.

Fig. 2: QKD Performance with respect to Distance



(a) Effect on Basis Match Rate

(b) Effect on QBER

Fig. 3: Impact of Increasing Z-Basis in the QKD Protocol

C. Evaluation Metrics

Evaluation records session success, QBER, transmitted pulses and acquisition rounds, reconciled key lengths, corrected errors, reconciliation leakage, privacy-amplified secret length and end-to-end latency. Protocol reliability is measured as

$$S_{\text{rate}} = \frac{N_{\text{successful}}}{N_{\text{total}}} \times 100 \tag{17}$$

where $N_{\text{successful}}$ represents the trials that successfully complete the required authentication, QKD, reconciliation, and key-establishment stages. The post-processing efficiency is characterized by the final secret-retention ratio

$$R_{\text{secret}} = \frac{L_{PA}}{L_{\text{sifted}}} \tag{18}$$

where L_{PA} and L_{sifted} denote the final privacy-amplified and initial sifted key lengths, respectively.

V. Results and Discussion

The OFAT study conducted shows that the proposed framework runs stably around the baseline operating region. The behaviour of the entire protocol also changes predictably as the baseline metrics change when the physical channel loss, fraction of Eve-intercepted channels, and protocol-specific parameters increase.

A. Effect of Communication Distance

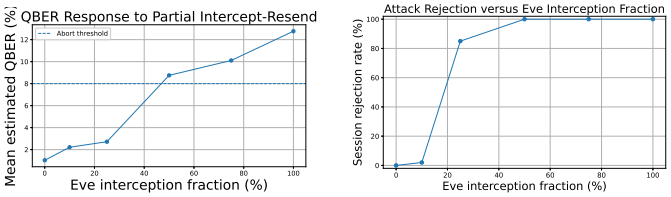
The physical layer was most strongly affected by the increase in communication distance. Without an Eve attack, the session success remained at 97-100% of channel success between 0-150 km (Fig. 2a). But under the same conditions, the channel success fell to 8% at 200 km distance while the mean transmitted pulse for channel completion increased from 3.09×10^4 at 0 km to 2.54×10^7 at 150 km and 2.43×10^8 at 200 km. The fibre attenuation exponentially increased the transmission requirements rather than the QBER. The mean QBER remains approximately 1% through 100 km before increasing to 1.29% at 150 km and 3.96% at 200 km as dark counts become non-negligible (Fig. 2b). The mean privacy-amplified secret bitstring length remains 1.28-1.36 kbits until 100 km, decreases slightly to 1.17 kbits at 150 km and collapses due to low session success at 200 km. These results show that 150 km is a practical upper operating region for the simulated baseline configuration.

B. Effect of Z-Basis Bias in the QKD

According to the T12 QKD protocol and the simulations results of Fig. 3a, it can be observed that increasing the Z-basis probability improves sifting efficiency, and the basis-match rate increases from 50.1% at $P(Z) = 0.5$ to 68.1% at the $P(Z) = 0.8$ baseline and 88.2% at $P(Z) = 0.9375$, with mean sifted-key lengths increasing from 2051 to 2788 and 3614 bits, respectively. However, overincreasing the bias reduces adversary detection. Under 100% interception, all sessions are rejected for $P(Z) \leq 0.85$, whereas approximately 5% succeeded at $P(Z) = 0.9375$. The mean estimated QBER simultaneously decreases from 25.6% under unbiased selection to 2.33% at the strongest bias. Thus, $P(Z) = 0.8$ provides a stable trade off between T12 sifting efficiency and reliable interception detection (Fig. 3b).

C. Adversarial Interception Behaviour

Increasing intercept-resend activity produces a clear security response. From Fig. 4a, it can be observed that the mean QBER increases from 1.03% without Eve to 2.21% at 10% interception and continues rising with the interception fraction. Consequently, session success decreased from 100% without Eve to 98% at 10%, 15% at 25%, and 0% for interception fractions of 50% or higher (Fig. 4b). Surviving secret material



(a) Effect on QBER (b) Effect on channel success

Fig. 4: Impact of Eve on QKD performance.

decreased simultaneously, with mean privacy-amplified key length falling from approximately 1312 bits without Eve to 729 bits at 10% interception and approximately 49 bits at 25%. Thus, interception is reflected not only in QBER-based rejection but also in sharply reduced session reliability and usable secret material.

VI. Conclusion and Future Work

This work presents an end-to-end quantum-secured biometric authentication and communication framework. It integrates local biometric reconstruction, ML-DSA-based device and user authentication, and a QAuth session authorisation verifier. Furthermore, T12 biased-basis QKD, recursive cascade reconciliation, and privacy amplification provide QKD information theoretic security. The HKDF-based key derivation and AES-GCM protect the entire communication channel. Each subsystem together provides a biometric verified secure channel without transmitting any reusable biometric templates in the process. Experimental evaluation demonstrated high legitimate-session reliability up to approximately 150 km under the validated baseline configuration, while revealing measurable security-efficiency trade-offs. In particular, aggressive basis bias increased sifting efficiency but weakened disturbance observability, whereas increasing adversarial interception progressively reduced session success and usable secret material. This simulation is an experimental prototype of an integrated QKD–PQC framework for secure iris-based biometric authentication. Future research can concentrate on replacing synthetic iris codes with real sensor data. It can also include scalable QKD networking through trusted or quantum-repeater-supported architectures, distributed key management, and continuous biometric authentication for periodically revalidating user presence during established sessions.

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